Mohd Talha Patrawala
CMPN-B
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ML: Assignment-7

Q.1.

1] ① Regularization:
ⓐ Controls tree complexity by penalizing large leaf weights.
ⓑ Prevents overfitting and improves generalization.

② Second-order Gradient optimization:
ⓐ Uses both first-order (gradient) and second-order (Hessian) information.
ⓑ makes more accurate split decisions.
ⓒ Leads to faster convergence and more stable learning.

2] Trade-offs:
① It will have more hyperparameters which requires proper tuning.
② Higher computational cost and complexity.
③ Less interpretable due to deeper trees and many boosting rounds.

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Q.2.

1] ① models make similar errors:

ⓐ If models are highly correlated, they may all misclassify the same samples.

ⓑ majority vote then reinforces the same mistake.

② Unequal model strengths:

ⓐ A strong model may get overridden by two weaker models.

ⓑ majority vote treats all models equally, ignoring accuracy differences.

2] ① Weighted Voting:

Assign higher weight to better models.

② Stacking:

Train a meta-model that learns how to combine predictions from base models.

③ Probability Averaging:

Instead of class vote, average predicted probabilities.

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